Minxing Zhang

Fourth-Year Ph.D. Candidate in Computer Science | Duke University
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Research focus. My research centers on privacy, generalizability, and reliable, cost-efficient evaluation for trustworthy LLMs and AI agents. My recent work explores reliable person distillation and representation, advertising for LLM agents, and question-asking capabilities in LLMs.

Education

Duke University 2023–Present
Ph.D. in Computer Science
Durham, NC | Adviser: Jian Pei

Emory University 2019–2022
B.S. in Computer Science and Applied Mathematics & Statistics
Atlanta, GA | GPA: 3.984/4.00; Highest Honor | Advisers: Li Xiong and Liang Zhao

Work and Internship Experience

Amazon Web Services | Applied Scientist Intern Summer 2026
Amazon Bedrock Guardrails, Deep Engine-Science | Seattle, WA *Research paper in submission*

Developed **Judge2Vec**, an unsupervised framework that selects compact, complementary LLM juries using criterion-specific competence embeddings estimated from leave-one-family-out agreement with a family-balanced consensus on unlabeled pilot examples.

Selected **4 of 19 judges**, ranking first among comprehensive baselines on 8 text benchmarks. Kendall correlation reached 0.467 vs. 0.424 for full-pool aggregation with **83.9% lower deployment cost**.

Enabled zero-shot jury transfer without target-task examples, retaining 98.5% of performance; added budget adaptation and extended evaluation to multimodal and agentic settings.

AT&T Chief Data Office | Graduate Research Intern Summer 2025
Bedminster, NJ | Mentors: Divesh Srivastava and Yaron Kanza *Research paper in submission*

Developed a kNN-LM framework for question answering and **selective forgetting without retraining the LLM or embedding function**. Designed embedding functions that preserve accurate next-token retrieval while protecting against leakage of deleted private associations: sensitive values are masked during embedding training, while correct next tokens remain retrievable from the private database until deletion.

Achieved **73.8% answer accuracy vs. 60.0%** for RAG.

Implemented FAISS retrieval and user-level logical deletion without index rebuilds; scaling the synthetic datastore 1,000-fold to **19.6 million key–value pairs** increased generation time only 4.9-fold (1.42–7.01 seconds/query).

Microsoft & Emory University | DataThink Research Project Summer 2022
Atlanta, GA

Built XGBoost rental-price prediction and SKATER spatial clustering models to analyze metro Atlanta's rental market.

Selected Research Projects

Person distillation. Formulated person distillation as modeling individual knowledge, judgment, and behavior from heterogeneous personal traces; established a lifecycle taxonomy and framework for evaluating person-specific fidelity.

LLM-based agent generalizability. Defined agent generalizability and organized evaluation and improvement methods by agent component; distinguished generalizable frameworks from agents that generalize to unseen settings.

MPCEval. Developed a reference-free, reproducible benchmark for multi-party conversation generation, with turn- and conversation-level metrics for speaker modeling, content quality, and speaker–content consistency. Benchmarked diverse generation methods across datasets spanning collaborative problem-solving, scripted dialogue, and real-world enterprise discussions. **KDD 2026 Datasets and Benchmarks Track, oral.**

Privacy attacks and defenses. Studied membership-inference attacks on teacher–student pairs and proposed mitigations for privacy leakage in knowledge distillation. Also developed a temporal attack on sequential data releases using hidden Markov models and reinforcement learning.

Earlier research at Emory. Three first-author papers on spatial interpolation, self-supervision, and trajectory generation.

Technical Skills

Languages: Python, Java, R, SQL | **Research tools:** LaTeX

Publications

Published and Accepted

[1] MPCEval: A Benchmark for Multi-Party Conversation Generation
Minxing Zhang[*], Yi Yang[*], Zhuofan Jia, Xuan Yang, Jian Pei, Yuchen Zang, Xingwang Deng, Xianglong Chen
ACM SIGKDD, Datasets and Benchmarks Track; **oral presentation**. [Paper] [Code] 2026

[2] Generalizability of Large Language Model-Based Agents: A Comprehensive Survey
Minxing Zhang, Yi Yang, Roy Xie, Bhuwan Dhingra, Shuyan Zhou, Jian Pei
ACM Computing Surveys, 58(10), Article 263, 44 pages. [Paper] 2026

[3] On Membership Inference Attacks in Knowledge Distillation
Ziyao Cui[*], **Minxing Zhang**[*], Jian Pei
ACM SIGKDD Explorations Newsletter, 28(1), pp. 32–40. [Paper] 2026

[4] Protecting Data Buyer Privacy in Data Markets
Minxing Zhang, Jian Pei
IEEE Internet Computing, 28(4), pp. 14–20. [Paper] 2024

[5] CSGAN: Modality-Aware Trajectory Generation via Clustering-Based Sequence GAN
Minxing Zhang, Haowen Lin, Shun Takagi, Yang Cao, Cyrus Shahabi, Li Xiong
IEEE MDM, pp. 148–157. [Paper] 2023

[6] Deep Spatial Prediction via Heterogeneous Multi-Source Self-Supervision
Minxing Zhang, Dazhou Yu, Yun Li, Liang Zhao
ACM Transactions on Spatial Algorithms and Systems, 9(3). [Paper] 2023

[7] Deep Geometric Neural Network for Spatial Interpolation
Minxing Zhang, Dazhou Yu, Yun Li, Liang Zhao
ACM SIGSPATIAL; Best Poster Runner-Up. [Paper] 2022

Preprints

[8] Person Distillation: Foundations and Principles for Personalized AI
Minxing Zhang[*], Yi Yang[*], Jian Pei
Preprints.org, 202608.1530.v1. [Paper] [Website] [Github] 2026

[9] Learning to Attack: Uncovering Privacy Risks in Sequential Data Releases
Ziyao Cui[*], **Minxing Zhang**[*], Jian Pei
arXiv:2510.24807. [Paper] 2025

Tutorials and Invited Talks

KDD | Tutorial Co-organizer 2026
Trustworthy LLM-Based Agents for Data-Centric AI. [Website]

PAKDD | Tutorial Co-organizer 2026
Generalizability of Large Language Model-Based Agents. [Program]

Samsung ML Reading Group | Invited Speaker 2024

Honors and Awards

Academic Excellence Award, Emory Computer Science 2023
Phi Beta Kappa, Gamma Chapter of Georgia 2023
Best Poster Runner-Up, ACM SIGSPATIAL 2022
Travel Grant, ACM SIGSPATIAL 2022